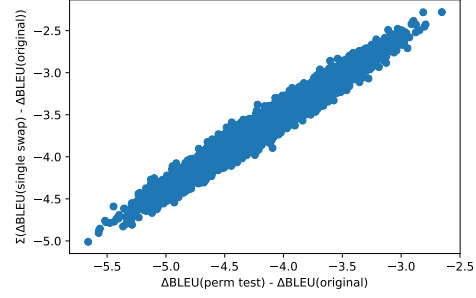
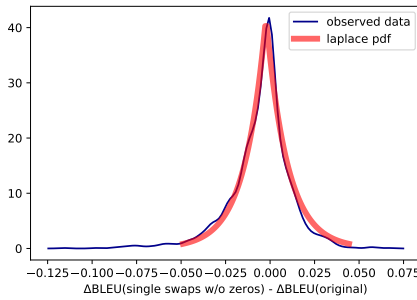


(a) Model trained on WMT19 data versus model trained on WMT18 data, evaluated on the 2019 test set.

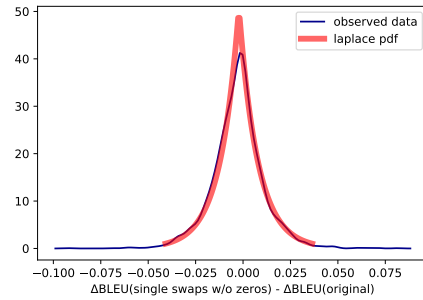


(b) Model trained on WMT18 data versus model trained on WMT16 data, evaluated on the 2018 test set.

Figure 15: Correlation between individual changes to Δ_B and the net effect.



(a) Model trained on WMT19 data versus model trained on WMT18 data, evaluated on the 2019 test set.



(b) Model trained on WMT18 data versus model trained on WMT16 data, evaluated on the 2018 test set.

Figure 16: Fitting a Laplace distribution to individual non-zero effects.

dividual amounts as the total effect of flipping a random subset of pairs. Based on the null distribution, we compute if the difference is significant for this trial. Repeating this many times and observing the proportion of trials that are found to be significant gives us the approximate power.

Figure 17 shows power for a range of values for Δ_B , n , P_0 and b_0 . When P_0 is low, as is true for the observed data in Table 3, effect sizes and sample sizes need to be larger in order for an experiment to be well-powered. But as P_0 gets higher, a given effect size can be detected by a smaller sample size. On the other hand, as b_0 increases and consequently the scale parameter b for the Laplace grows, even large effect sizes cannot be detected by test sets containing 5,000 samples.

H Details of Human Evaluation Section

H.1 Meta-analysis of human ratings for EMNLP 2019

To assess the state of statistical power in a typical NLP study using human evaluation, we sampled papers from the mean EMNLP 2019 workshop that

contained the phrase “human eval”. This first pass returned 117 papers, of which 86 had relevant human evaluations (in which models were compared), with the remainder either referencing human evaluation, or containing some other type of evaluation, such as comparing the agreement between automated metrics and human performance. Because some papers had more than one such evaluation, we had 97 experiments for analysis. Of these 51 were Likert experiments (as discussed in the main text), 38 were some form of direct model comparison, and 8 were other.

Significance testing was rare and was reported, in some form, in only 24% of experiments. Bold-ing or starring the best results in a table was more common, occurring in 63% of human rating experiments in our set. Whether bold results implies that the author is claiming a meaningful difference is not always clear. We did find one single case of authors performing a power analysis to estimate sample size among the papers we surveyed (Garbacea et al., 2019). However, because that paper did not involve a comparison of models to a baseline, it

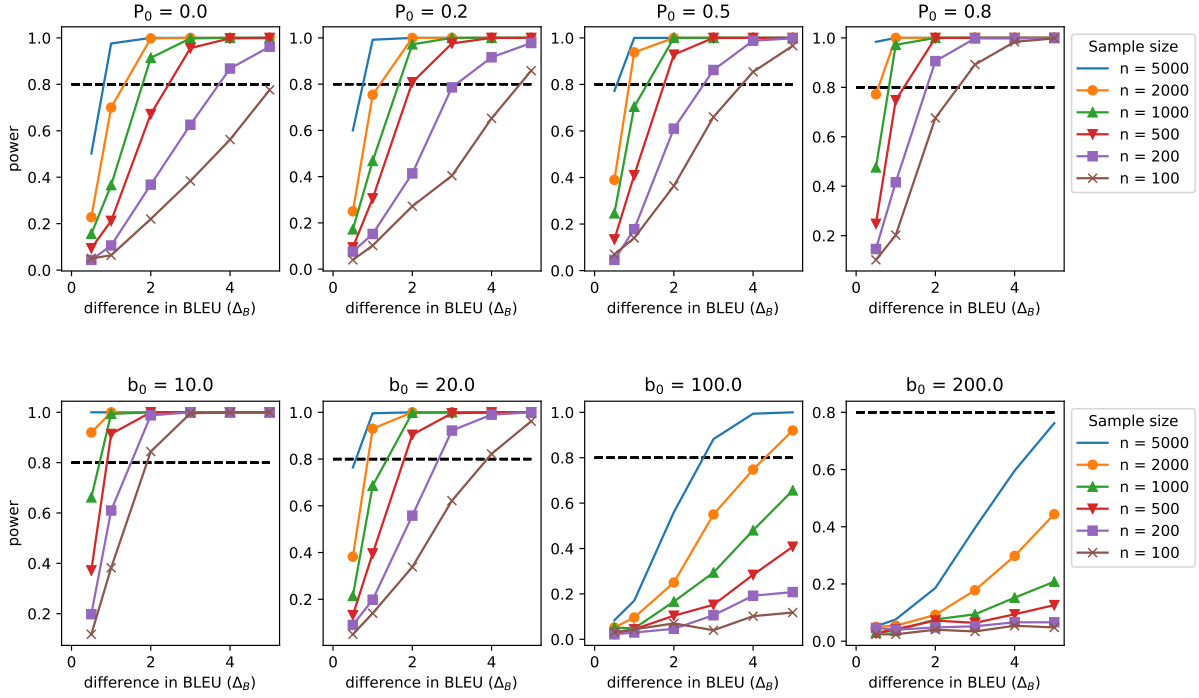


Figure 17: Power Analysis for BLEU scores: Variation in estimates of power for different values of P_0 (top) and b_0 (bottom). For the top row, $b_0 = 25.8$, and for the bottom row, $P_0 = 0.13$.

was not included in our analysis. In addition, we note that few details were provided, such that we were unable to ascertain precisely how the power analysis was done.

Because we chose to focus on ordinal ratings, we further annotated those in order to record the mean ratings and experimental characteristics (number of annotators, number of items, number of annotators per item), as well as all differences for all metrics between the model being proposed and the best performing baseline evaluated in the paper, as discussed in the main text.

H.2 Human evaluation datasets

For our analyses, we make use of the following datasets:

- From Hashimoto et al. (2019) we use the evaluation data for Reddit, language modeling, and summarization. The data is available at <https://worksheets.codalab.org/worksheets/0x88644b5ee189402eb19d39d721d1005c>
- From Dathathri et al. (2020) we use the available ratings. The data is available at <https://github.com/uber-research/PPLM>
- For WMT19 (<http://statmt.org/wmt19/>

[translation-task.html](#)), the data is available at <https://www.computing.dcu.ie/~ygraham/newstest2019-humaneval.tar.gz>

- For Holtzman et al. (2020), we obtain the human evaluation data directly from the authors.

H.3 Linear Mixed Effect Models

To assess power in the human ratings framework, we used linear mixed effect models with random intercepts and slopes for worker and item, as in Barr et al. (2013). Following best practices, we use the following structure, where w is a particular worker and i is a particular item. There are seven parameters, corresponding to the parameters needed for running a power analysis: fixed effects β_0 (the intercept) and β_1 (the model effect), and variance parameters for the worker intercept (σ_{0w}), the item intercept (σ_{0i}) and their respective slope variance parameters (σ_{1w} and σ_{1i}). There is also a variance parameter for the overall error (σ_{wi}). We transform the Likert ratings to be on a $[0, 1]$ scale and treat them as normally distributed (which we note is an imperfect assumption). We give fit parameters for these values, on a few datasets, in Tables 13, 14, and 15.

$$Y_{wi} = \beta_0 + W_{0w} + I_{0i} + (\beta_1 + W_{1w} + I_{1i})X_i + e_{wi} \quad (5)$$

$$I_{0i} \sim N(0, \sigma_{0i}) \quad (6)$$

$$W_{0i} \sim N(0, \sigma_{0w}) \quad (7)$$

$$I_{1i} \sim N(0, \sigma_{1i}) \quad (8)$$

$$W_{1i} \sim N(0, \sigma_{1w}) \quad (9)$$

$$e_{wi} \sim N(0, \sigma_{wi}) \quad (10)$$

For simplicity and convergence issues, we do not include a correlation parameter in the random effect structure.

To assess power, we use two possible variance settings derived from the model fits (“high variance” and “low variance” settings, in the main text) and show these in Table 16. We systematically vary the number of annotators (always assuming each annotator annotates each item, which is not always true in typical experiments), the number of items, and the effect size. We note that simulations can be customized to the planned analysis, including aspects such as how many items will be annotated by each annotator.

To compute power, we use each setting of the parameters to simulate 200 experiments and compute the proportion that detect a significant positive effect ($t > 1.96$). Significant effects in the opposite direction ($t < -1.96$) do not count as detections. Code for these model fits and simulations is included with the online materials. However, we note that these should be used as a starting point, rather than being blindly copied, as details may differ in each experimental setting.

H.4 Head to head human evaluations

Another commonly used form of human evaluation is head to head comparison, where raters are shown a pair of outputs (one from each model), and asked to choose which they prefer, sometimes with “neither” as a third option. Head to head comparisons offer some advantages over ratings-based approaches (Yannakakis and Martínez, 2015; van der Lee et al., 2019), but do not scale as well when comparing many models.

As with ordinal judgements, there are multiple ways of analyzing such data. If we treat annotator judgements as independent and identically distributed (such as if we only collect one judgement from each annotator), we could model this simply in terms of the underlying probabilities that

a random annotator will prefer each model (as in the opening example in the main paper). In that case, running a power analysis would be a simple as assuming values for the underlying probabilities of each category (win, lose, draw), as usual based on pilot data or prior assumptions, and simulating many draws from that prior, checking in each sample to see if there is a statistically significant difference between win and lose.

On the other hand, if multiple judgements will be collected from each annotator and/or for each pair of outputs, then it makes sense to use a richer model to account for all sources of variation, as described above (see §H.3). In particular, the mixed effects framework can be adopted, potentially by modeling the outcome as a logistic model (in the case of win or lose), with ties either excluded or split.